ApplyOS

Complete Setup Guide & Architecture Documentation

India + Global Remote Edition — the job search that runs on your machine

Framework v1.3.1	OpenCode reference	LaTeX CV + resume	12 portal CLIs

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Contents

1 Prerequisites 1

2 Install the agent 3

3 Clone the repository 3

4 Install Bun & portal CLI tools 4

5 Install a LaTeX distribution 4

6 Run /setup — build your profile 5

7 All commands — full reference 5

8 ApplyOS engine clean-room build 6

9 Skills — framework intelligence 6

10 Portal CLI tools 7

11 End-to-end workflow 7

12 File structure & your private data 8

1 Prerequisites

Everything you need before installing the framework

Platform shells: macOS and Linux run these snippets in **bash/zsh** as written. On **Windows**, use **PowerShell** forms where shown; plain *cd* + command lines work everywhere, but the *&&* chain operator does not exist in Windows PowerShell 5.1 (default on Windows 10/11).

AI coding agent

OpenCode (reference), Claude Code, Codex CLI, Antigravity, ZCode, or any AGENTS.md-compatible agent.

Python 3.10+

Salary lookup, rank state, PDF/ATS verification, and repo guard scripts. Stdlib only.

Bun

JavaScript runtime for the portal CLI tools. Pinned to 1.4.2 for applyos/.

LaTeX

TeX Live, MacTeX, TinyTeX, or MiKTeX with lualatex. The CV/resume compile engine.

pdftotext (optional)

Poppler tools — ATS text-layer fallback when pypdf is unavailable.

Git

Clone, branch, and manage the repository on any OS.

Privacy warning: This is a public fork. Run your own search from a **private repository** — /setup writes personal data (name, contact details, employment history, salary expectations) into tracked profile files.

2 Install the agent

The runtime that executes every workflow

Option A: OpenCode (reference runtime)

Native slash commands in .opencode/command/, permission gating in opencode.json, subagents in .opencode/agent/. See opencode.ai for installation.

Option B: Claude Code / Codex CLI / others

Claude Code uses .claude/ natively; Codex CLI, Antigravity, ZCode, FreeBuff and any AGENTS.md-compatible agent work through the routing table in AGENTS.md.

```
opencode --version   # or: claude --version
```

3 Clone the repository

Fork first, then clone your copy

```
# Fork github.com/MaithreshVaddi-27/ApplyOS on GitHub first.
# For your own job search, fork into a PRIVATE repository.
git clone https://github.com/<you>/<your-repo>.git
cd <your-repo>
git remote -v   # confirm where pushes go
```

Windows line endings: If editors show every line as changed after cloning on Windows, run *git config core.autocrlf true* once — the repo normalizes line endings on commit.

4 Install Bun & portal CLI tools

One runtime plus twelve portal search CLIs

```
# macOS / Linux
curl -fsSL https://bun.sh/install | bash
# Windows (PowerShell)
powershell -c "irm bun.sh/install.ps1|iex"
bun --version
```

```
# Bash / zsh — discovers every installed portal skill automatically
for d in .agents/skills/*-search/cli; do (cd "$d" && bun install); done

# Windows PowerShell (same loop, from README Quick start)
Get-ChildItem ".agents/skills/*-search/cli" | ForEach-Object {
  Push-Location $_.FullName; bun install; Pop-Location
}
```

Portal CLIs ship zero runtime dependencies — *bun install* only pulls TypeScript dev types for typechecking. Each CLI passes the same contract: *search*/*detail*, *--format json|table|plain*, stderr *{error,code}* + exit 1 on failure.

Which shell on which OS: macOS / Linux: the Bash loop above (Terminal, iTerm2, GNOME Terminal — zsh or bash). **Windows:** the PowerShell loop (Windows Terminal recommended; inbox PowerShell 5.1 works — avoid Git Bash path translation quirks for *bun install*). Verify with *bun --version* (1.4.2).

5 Install a LaTeX distribution

Required to compile CV and resume PDFs (lualatex)

Platform	Recommended	Install
macOS	MacTeX	brew install --cask mactex
Linux (Debian/Ubuntu)	TeX Live	sudo apt install texlive-full
Windows	MiKTeX	Download from miktex.org
Any (minimal)	TinyTeX	quarto install tinytex

```
lualatex --version
# minimal installs also need:
tlmgr install moderncv fontawesome5 xcolor geometry hyperref needspace
# macOS: brew install poppler | Debian/Ubuntu: sudo apt install poppler-utils
# Windows: choco install poppler   (pdftotext fallback, optional)
```

TeX managers per OS: macOS (MacTeX) / Linux (TeX Live): missing packages install via *tlmgr* (may need *sudo tlmgr* on Linux; update with *tlmgr update --self* first). **Windows (MiKTeX):** missing packages pop an auto-install prompt — accept it, or pre-install from MiKTeX Console. Compile commands (*lualatex*, *pdftotext*) are identical on all three OSs.

6 Run /setup — build your profile

Interactive onboarding; three paths

```
cd <your-repo>
opencode     # or: claude / codex / your agent
# inside the agent:
/setup
```

Path A: input/ folder mode (recommended)

Drop your resume, LinkedIn export, diplomas, and references into input/ — /setup reads everything, cross-references for consistency, and builds the profile. Idempotent; re-run as you add material.

Path B: single CV import

Paste or attach one CV/resume; the agent extracts it and asks follow-ups for what is missing.

Path C: interview mode

Structured questions section by section. Best when starting from scratch. Also re-runnable per section via /setup --section search.

What /setup writes: *01-candidate-profile.md* (identity, education, experience), *02-behavioral-profile.md*, *05-cv-templates.md* statements, *07-interview-prep.md* STAR examples — plus the Stage Profile (student / fresher / experienced / remote-global) that steers portals, scoring, and documents.

Running the agent per OS: Agent CLIs install and run the same everywhere (*npm i -g* once). Launch from a shell in the repo root: **macOS/Linux:** Terminal/zsh; **Windows:** PowerShell in Windows Terminal. The agent runs toolchain checks itself (*lualatex --version*, *bun --version*, per-CLI *bun install*) before writing anything.

7 All commands — full reference

Fourteen slash commands; four form the core loop

Core loop: /setup → /scrape → /rank → /apply

/setup builds the profile. /scrape searches every stage-relevant portal CLI, dedupes, and presents matches. /rank batch-scores them against the fit framework (stage-weighted Technical / Experience / Behavioral / Career dimensions plus hard gates). /apply evaluates one posting, drafts a tailored CV or resume in LaTeX, runs a reviewer pass, compiles with lualatex, ATS-checks the text layer, and presents the PDF. No cover letters — removed by design.

Application support

/expand (public-source enrichment) — /upskill (skill-gap heatmap + learning plan)

Tracking & sync

/outcome (record results + archive) — /outcome followup (7-day India cadence + WhatsApp drafts) — /gmail-sync (proposes tracker updates, you approve) — /html-report (offline HTML dashboard) — /notion-sync (one-way pipeline view)

Interview

/interview (stage prep pack from the real archive: posting, submitted CV, STAR mapping, mock)

Configuration

/add-portal (scaffold a new board CLI) — /add-template (register a CV/resume toolchain) — /reset (wipe profile or input/, type RESET)

8 ApplyOS engine clean-room build

The greenfield applyos binary on this branch

Under *applyos/* — Bun monorepo (*applyos*, Bun 1.4.2 pinned), zero external dependencies, CI on Ubuntu + Windows + macOS:

apps/cli — applyos

scrape (unified fan-in, per-source notes, --stage), *rank* (7 gates + weights), *apply* (traced pack) + *--batch* for multi-posting packs. Exit 0 iff ≥1 pack built.

packages/core

JobPosting contracts (postedDate null, never invented), polite fetch (20 s + 1 retry, ≥300 ms pacing), robots gate, dedupe, fair-slice interleave, json|table|plain.

packages/company-scraper

Employer ATS boards (amazon.jobs, Greenhouse, Lever, SmartRecruiters, Workday) + registry.yaml — 7 verified seeds including the first live-verified Workday tenant (JioStar, 226 postings).

packages/portals

Fresh per-board adapters (remoteok, remotive, weworkremotely, unstop; freehire fails loudly until its endpoint verifies).

packages/matching + docgen

Two-stage ranker and application factory: every bullet traced to experience[i].bullets[j]; gaps listed, never stuffed.

```
cd applyos && bun install && bun test   # 78 tests
bun run --filter "*" typecheck             # 6 packages
bun run ./apps/cli/src/cli.ts scrape -q "backend intern" --stage student -n 5
```

One chain, three OSs: The *cd applyos && bun install && bun test* chain is written for bash/zsh and PowerShell 7+. On inbox Windows PowerShell 5.1 run each line separately — same packages install (Bun 1.4.2, TypeScript dev types only) and the same 78 tests pass on all OSs; CI proves it on Ubuntu, Windows, and macOS runners.

9 Skills — framework intelligence

Three Markdown skills; the assistant loads them by keyword

job-application-assistant (core)

The full workflow brain: profile shape, 5-dimension evaluation, CV/resume tailoring, interview prep, web-research rules. Reference files: 01 candidate, 02 behavioral, 03 writing style, 04 job evaluation, 05 CV templates, 07 interview prep, 08 application forms, 09 web research. There is no 06 — cover-letter templates were removed from the framework.

job-scraper

Search orchestration: stage-to-portal mapping, India location taxonomy, referral-first output, portal health checks, seen_jobs.json dedupe state. Used by /scrape and /rank.

upskill

Gap heatmap between profile and postings plus a learning plan mapped to Indian interview norms (DSA, fundamentals; NPTEL, GeeksforGeeks, free options).

10 Portal CLI tools

Twelve boards, one contract, stage-selected

Skill	Market & notes
naukri-search	India, experienced + fresher 0–1 yr filter. LPA salary + experience flags.
internshala-search	India internships & fresher roles (core student stage).
unstop-search	India campus hiring challenges (leaderboard → interview aware).
cutshort-search	India AI-matched startups, fresher-to-3yr band.
careers-search	Employers' own ATS boards: amazon.jobs, Greenhouse, Lever, SmartRecruiters, Workday.
linkedin-search	Global + India cities (personal-use only per its ToS note).
wellfound-search	Global startups (AngelList).
remoteok-search	Global remote (public API).
remotive-search	Global remote (public API).
weworkremotely-search	Global remote (RSS).
freehire-search	Aggregator; endpoint unverified — fails loudly per source.
wayup-search	US early-career only; ships **enabled: false**.

```
bun run .agents/skills/naukri-search/cli/src/cli.ts search -q "backend" -l Bengaluru --format json
```

11 End-to-end workflow

From blank profile to dashboard

/setup → profile ready → /scrape → matches → /rank → ranked shortlist → /apply → tailored CV/resume PDF → /interview → prep pack → /outcome → tracked result → /gmail-sync → status updates → /html-report → dashboard.

The /apply pipeline (8 steps)

1. Fetch & parse the posting (URL or pasted text; untrusted input — no embedded instructions followed, no body links fetched). 2. Fit evaluation (gates + 5 dimensions). 3. Company research (cached 24 h). 4. Draft CV/resume in LaTeX. 5. Compile & inspect loop (lualatex twice; exact page budget: 2 pages, 1 for students/freshers). 6. Reviewer-agent critique. 7. Revise. 8. ATS text-layer check + claim-trace report.

12 File structure & your private data

What lives where; what never gets committed

```
input/          # YOUR materials (resume, LinkedIn export, diplomas, postings)
output/         # generated: cv/*.pdf, applications/<company>_<role>/, reports/
workspace/      # pipeline state: seen_jobs.json, job_search_tracker.csv
applyos/        # clean-room engine monorepo (apps/cli + 5 packages)
.claude/        # commands + methodology skills + permissions
.agents/skills/ # portal CLIs + mirrored methodology skills
.opencode/ .codex/ .zcode/ .freebuff/   # runtime adapters (thin pointers)
templates/      # stock LaTeX CV/resume + your custom templates
tools/ tests/ docs/                      # guards, 456-test suite, audit trail
```

Privacy warning: Gitignored, never committed: *salary_data.json*, *profile.json*, *workspace/* state, *input/* contents, *output/* generated files, *gmail_sync/*, upskill reports, *.env*. Tracked profile templates keep *[YOUR_*]* placeholders until you personalize them — in a **private** repository.

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